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Kelas : TI-25-PA2

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<https://github.com/252310057-boop/algoritma-semster2.git>

tugas 1

```
Isi Stack (Top -> Bottom):
124
94
201
999
621

-----
Process exited after 0.2378 seconds with return value 0
Press any key to continue . . .
```

```
1  #include <iostream>
2  using namespace std;
3
4  #define MAX 5
5
6  class Stack {
7  private:
8      int data[MAX];
9      int top;
10
11  public:
12      Stack() {
13          top = -1;
14      }
15
16      void push(int x) {
17          if (top == MAX - 1) {
18              cout << "Stack penuh!\n";
19          } else {
20              data[++top] = x;
21          }
22      }
23
24      void print() {
25          cout << "Isi Stack (Top -> Bottom):\n";
26          for (int i = top; i >= 0; i--) {
27              cout << data[i] << endl;
28          }
29      }
30  };
31
32  int main() {
33      Stack s;
34
35      s.push(621);
36      s.push(999);
37      s.push(201);
```

```

38     s.push(94);
39     s.push(124);
40
41     s.print();
42
43     return 0;
44 }

```

Tugas 2

```

Data dari Head ke Tail : 24 10 33 33
Data dari Tail ke Head : 33 33 10 24

-----
Process exited after 0.2899 seconds with return value 0
Press any key to continue . . .

```

```

1  #include <iostream>
2  using namespace std;
3
4  struct Node {
5      int data;
6      Node* next;
7      Node* prev;
8  };
9
10 class DoublyLinkedList {
11 private:
12     Node* head;
13     Node* tail;
14
15 public:
16     DoublyLinkedList() {
17         head = NULL;
18         tail = NULL;
19     }
20
21     void push(int data) {
22         Node* baru = new Node;
23         baru->data = data;
24         baru->next = NULL;
25         baru->prev = NULL;
26
27         if (head == NULL) {
28             head = tail = baru;
29         } else {
30             tail->next = baru;
31             baru->prev = tail;
32             tail = baru;
33         }
34     }
35
36     void tampilMaju() {
37         Node* bantu = head;

```

```

38 while (bantu != NULL) {
39     cout << bantu->data << " ";
40     bantu = bantu->next;
41 }
42 cout << endl;
43 }
44
45 void tampilMundur() {
46     Node* bantu = tail;
47     while (bantu != NULL) {
48         cout << bantu->data << " ";
49         bantu = bantu->prev;
50     }
51     cout << endl;
52 }
53 };
54
55 int main() {
56     DoublyLinkedList dll;
57
58     dll.push(24);
59     dll.push(10);
60     dll.push(33);
61     dll.push(33);
62
63     cout << "Data dari Head ke Tail : ";
64     dll.tampilMaju();
65
66     cout << "Data dari Tail ke Head : ";
67     dll.tampilMundur();
68
69     return 0;
70 }

```